

Customer Segmentation & Churn Pattern Analytics in European Banking

Step 1: Purpose of the project:

The main purpose of the project is to analyse the data of European bank and determine which customer is likely to leave the bank based on the factors like demography, age, credit policy, account balance, income, etc

Specific objective of the project:

Primary Objectives of the Project

1. To measure the overall customer churn rate in European banking.
2. To identify customer segments that are more likely to churn based on demographic, geographic, and financial characteristics.
3. To analyze churn patterns across different European regions (France, Germany, and Spain).
4. To understand the relationship between customer behavior factors such as tenure, activity level, and product usage with customer churn.
5. To develop an interactive Streamlit dashboard for visualizing customer churn patterns and segmentation insights.

Secondary Objectives of the Project

1. To analyze churn differences based on age groups and gender.

2. To evaluate churn behavior among customers with different credit score levels.
3. To study the financial profile of churned customers using balance and estimated salary data.
4. To identify high value
- 5.To support strategic decision-making through segmentation-based churn analytics.
- 6.To generate data-driven insights and recommendations for improving customer retention strategies.
- 7.To support strategic decision-making through segmentation-based churn analytics.

Step 2: Gathering and Preparing Data

Data Collection / Data Gathering

The dataset used for this project was collected from a customer banking dataset containing information about customers from European countries such as France, Germany, and Spain. The dataset includes customer demographic, financial, and behavioral information that helps in analyzing customer churn patterns.

The dataset consists of multiple customer attributes including:

- Customer ID
- Credit Score
- Geography
- Gender
- Age

- Tenure
- Account Balance
- Number of Products
- Credit Card Ownership
- Active Membership Status
- Estimated Salary
- Customer Churn Status (Exited)

The target variable used in this project is Exited, where:

- 0 = Customer Retained
- 1 = Customer Churned

The dataset was imported into Python using the Pandas library for further analysis and preprocessing.

Dataset Import Process

The following steps were followed to gather the data:

1. The CSV dataset file was downloaded and stored locally.
2. The dataset was loaded into Python using Pandas.
3. Initial inspection was performed to understand dataset structure.
4. The number of rows and columns was verified.
5. Data types of all variables were checked.

Basic inspection methods used:

- Viewing first few records
- Checking dataset dimensions

- Understanding column information
- Statistical summary of numerical variables

This helped in understanding the quality and structure of the dataset before performing analysis.

Data Preparation

After gathering the dataset, data preparation was performed to make the data suitable for customer churn analysis.

Data Inspection

Initially, the dataset was examined to understand:

- Total number of records
- Missing values
- Duplicate values
- Data types
- Unique categories in categorical columns

The dataset was checked to ensure that the customer churn label (Exited) was properly represented.

Removing Irrelevant Features

Certain columns were removed because they did not contribute to churn analysis.

The following non-analytical columns were excluded:

- CustomerId – Unique identifier only

- Surname – Customer last name, not useful for analysis
- RowNumber (if available) – Index column without analytical significance

These fields were removed because they do not influence customer churn behavior and may introduce unnecessary complexity.

Missing Value Handling

The dataset was checked for missing values using appropriate validation techniques.

If missing values were found:

- Numerical values would be treated carefully using mean or median methods.
- Categorical values would be handled using mode replacement.

However, since the dataset contained minimal or no missing values, extensive missing-value treatment was not required.

Duplicate Record Validation

Duplicate customer records were checked to ensure data quality and avoid repeated observations that could affect churn analysis.

Any duplicate rows found would be removed to maintain consistency in analysis.

Data Type Verification

Data types were validated to ensure proper analysis.

Examples:

- Numerical variables:
 - CreditScore
 - Age
 - Balance
 - EstimatedSalary
- Categorical variables:
 - Geography
 - Gender
- Binary variables:
 - Exited
 - HasCrCard
 - IsActiveMember

This ensured accurate grouping and visualization during exploratory analysis.

Creating Customer Segments

To better understand churn behavior, customer segmentation variables were created.

1. Age Segmentation

Customers were grouped into age categories:

- Young Customers (<30 years)
- Adult Customers (30–45 years)
- Middle-Aged Customers (46–60 years)
- Senior Customers (>60 years)

This helped in identifying which age groups showed higher churn.

2. Credit Score Segmentation

Customers were categorized based on credit score:

- Low Credit Score
- Medium Credit Score
- High Credit Score

This segmentation was used to evaluate financial stability and churn risk.

3. Tenure Segmentation

Customers were grouped according to years spent with the bank:

- New Customers
- Mid-Term Customers
- Long-Term Customers

This helped analyze whether customer loyalty influenced churn.

4. Balance Segmentation

Customers were categorized into:

- Zero-Balance Customers
- Low-Balance Customers
- High-Balance Customers

This allowed the analysis of churn among financially valuable customers.

Data Validation

Before moving to analysis, the prepared data was validated to ensure:

- No missing critical fields
- Correct churn labeling
- Proper customer segmentation
- Consistent binary variable values

After successful validation, the dataset became ready for Exploratory Data Analysis (EDA) and churn pattern analysis.

Step 3: Exploratory data analysis

Introduction to EDA

Exploratory Data Analysis (EDA) was performed to understand customer churn patterns in European banking. The analysis helped identify how churn varies across geography, demographics, customer engagement, and financial characteristics. Various visualizations and segmentation techniques were used to extract meaningful insights from the dataset.

Dataset Overview

Before performing analysis, the dataset was inspected to understand its structure.

The following checks were performed:

- Number of rows and columns
- Data types of variables
- Missing value verification
- Duplicate record checking
- Statistical summary of numerical variables

Include Screenshot Here:

```
df.info()
```

Interpretation:

The dataset contains customer demographic, financial, and behavioral information. Variables such as age, balance, credit score, geography, and activity status are available for churn analysis. The target variable used for analysis is Exited, which indicates whether a customer left the bank.

```
df.describe()
```

Interpretation:

The statistical summary provides an understanding of customer characteristics such as average age, average balance, salary range, and tenure. This helps identify the general customer profile within the banking dataset.

Data Cleaning and Preparation

Data preprocessing was performed to improve analysis quality.

The following actions were taken:

- Removed unnecessary columns:
 - CustomerId
 - Surname
 - RowNumber
- Verified missing values
- Checked duplicate records
- Created segmentation groups for age, credit score, tenure, and balance.

Missing values output (df.isnull().sum())

Interpretation:

No significant missing values were found in the dataset, indicating good data quality for analysis.

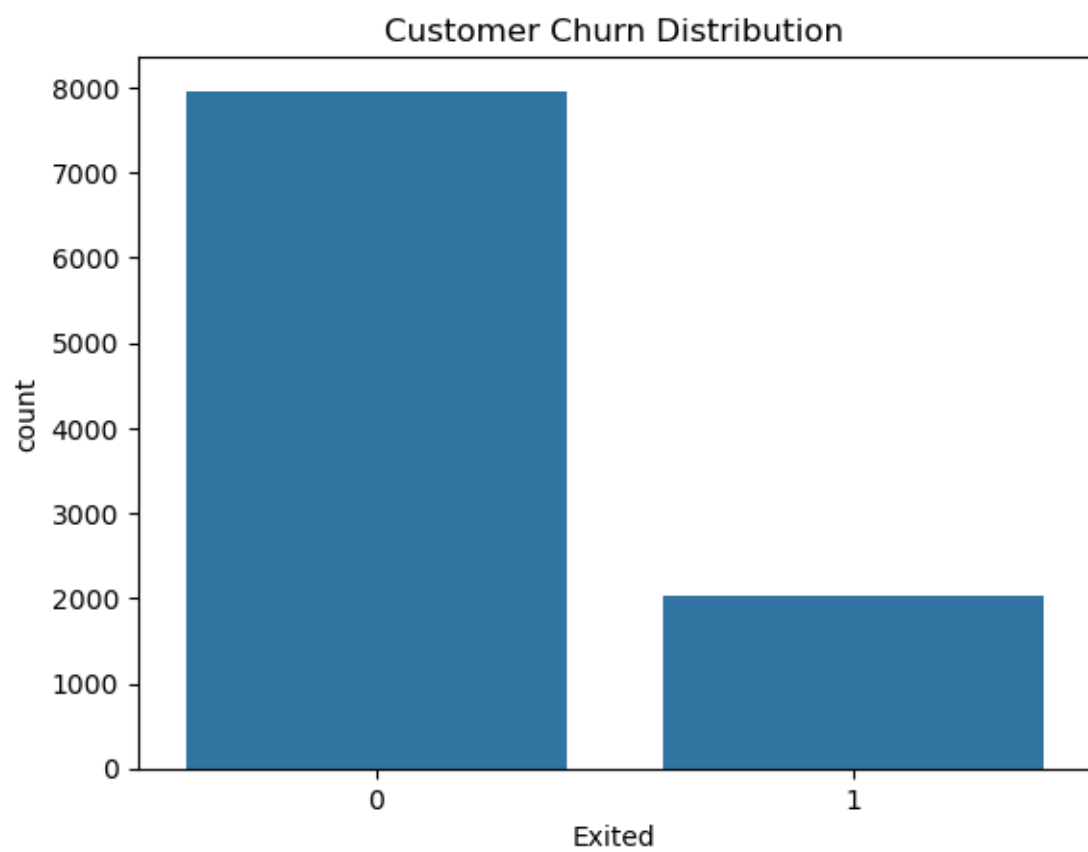
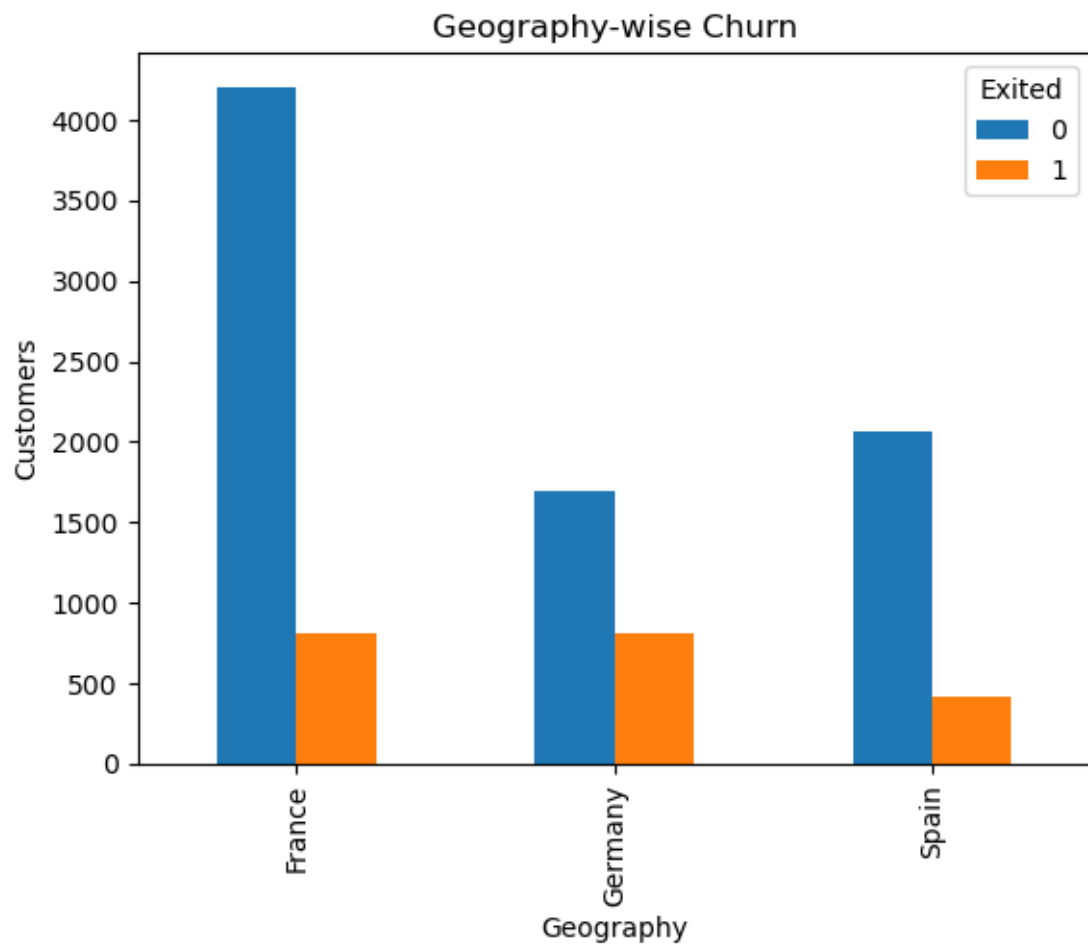
```
(df.duplicated().sum())
```

Interpretation:

The dataset was checked for duplicate customer records to ensure accurate churn analysis.

Overall Customer Churn Analysis

The overall churn rate was calculated to determine the percentage of customers who left the bank.



Interpretation:

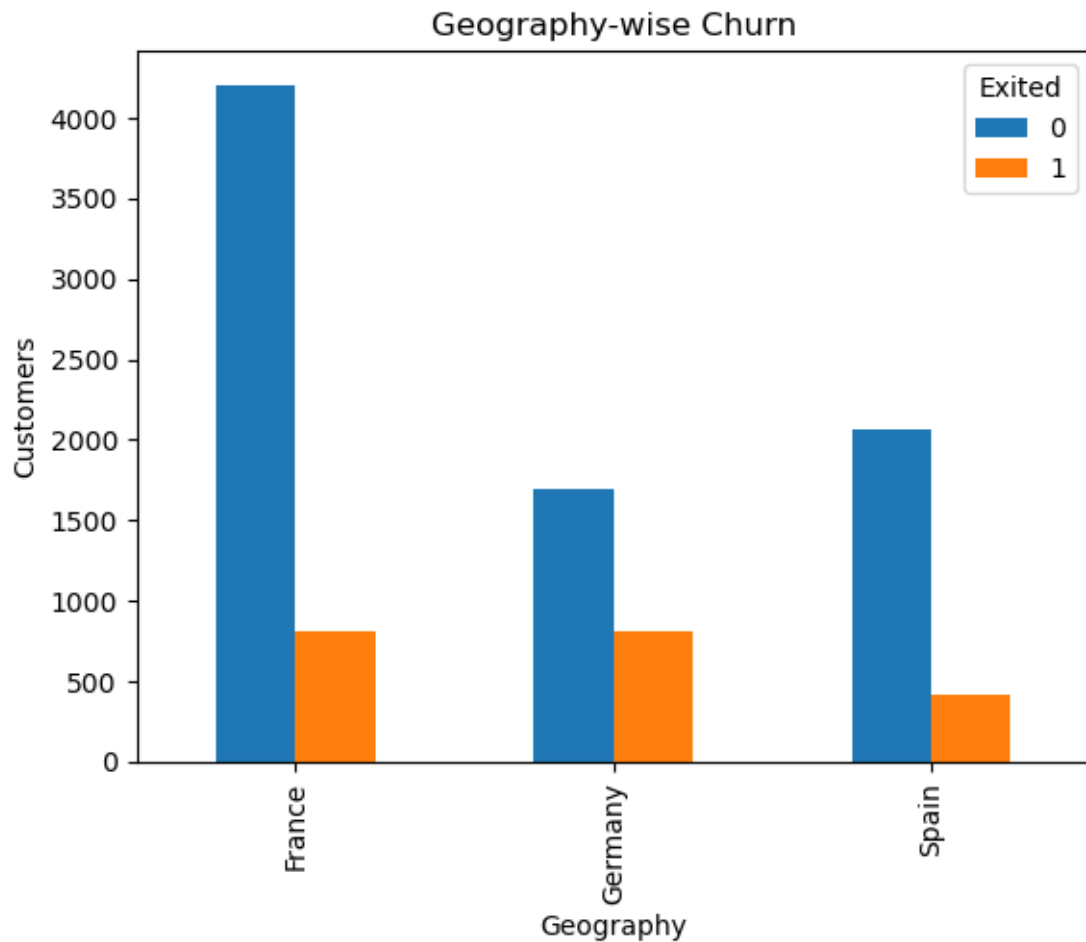
The visualization shows the distribution of retained and churned customers. A smaller portion of customers exited the bank, while the majority remained active. This indicates that customer churn exists but is concentrated among specific customer segments.

Geography-wise Churn Analysis

Customer churn was analyzed across different countries.

Countries included:

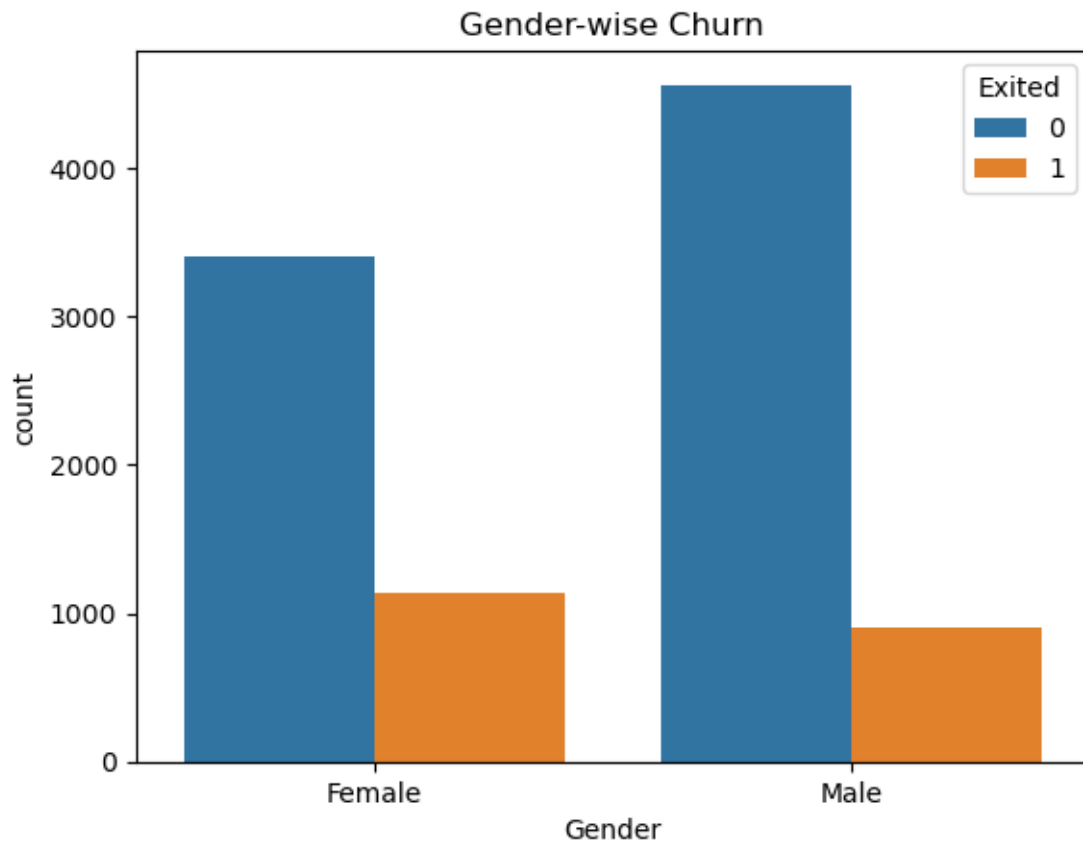
- France
- Germany
- Spain



Interpretation:

The graph compares churn behavior among different countries. Certain regions exhibit higher churn rates, indicating possible regional differences in customer satisfaction, competition, or banking services.

As showed in the graph France And Germany has higher churn rate compared to Spain



Gender-wise Churn Analysis

Churn behavior was analyzed based on gender.

Interpretation:

The analysis helps determine whether churn behavior differs between male and female customers. This provides insight into customer retention trends across gender groups.

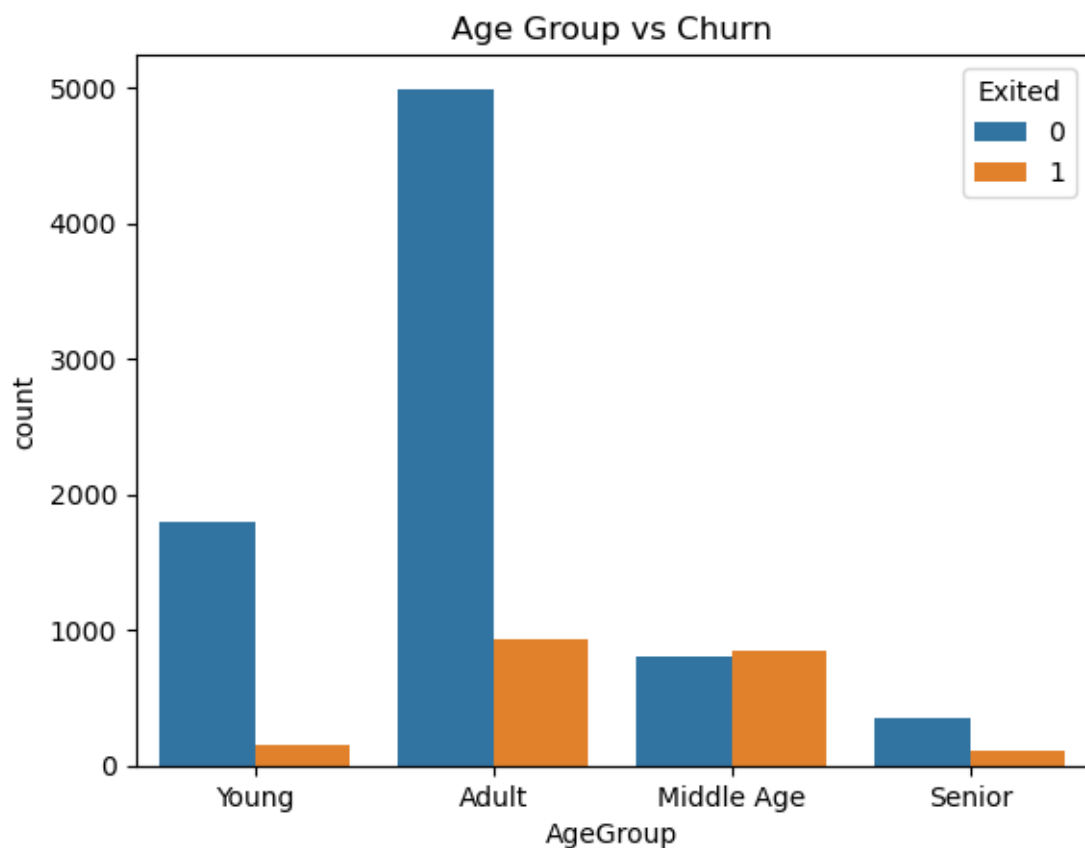
As observed female customers exited more than male

Age Group-wise Churn Analysis

Customers were segmented into age categories to understand age-related churn behavior.

Age Groups:

- Young
- Adult
- Middle Age
- Senior

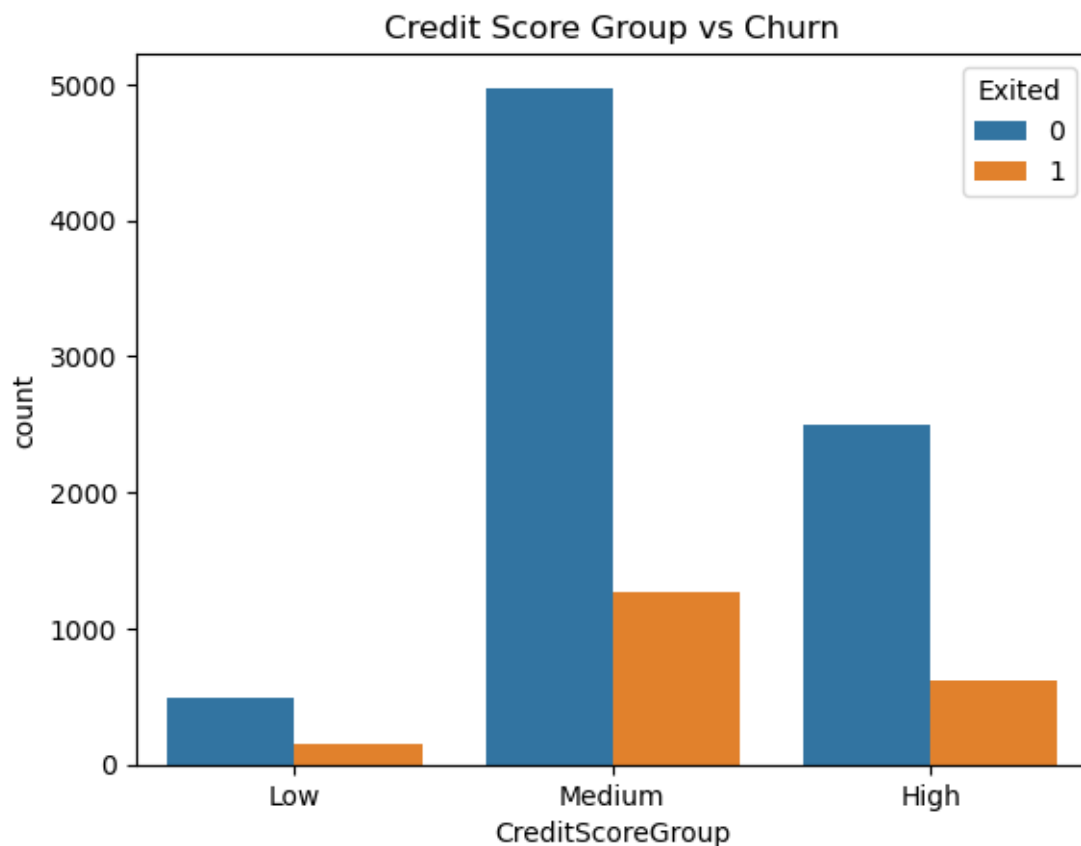


Interpretation:

The analysis identifies which age groups are more likely to leave the bank. Certain age categories may exhibit higher churn due to financial needs, service expectations, or lifestyle changes. Middle-aged customers showed higher churn than younger customers.

Credit Score-wise Churn Analysis

Customers were segmented according to credit score levels.



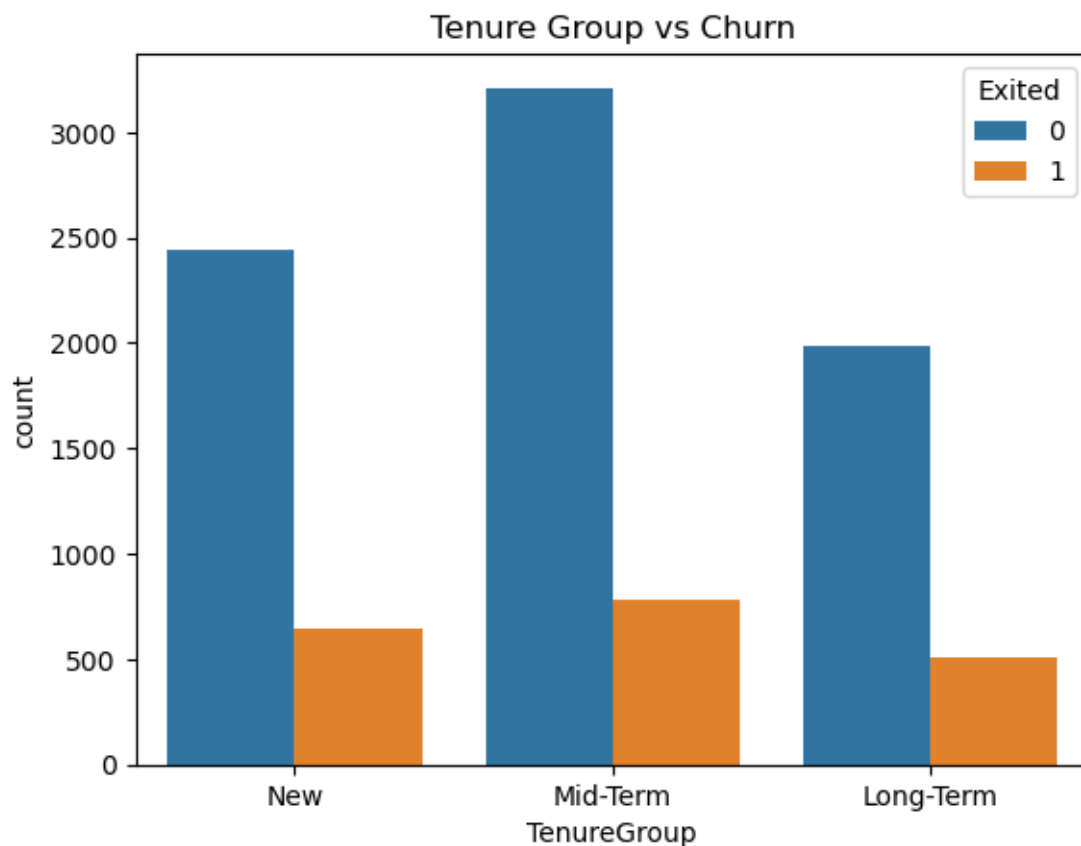
Interpretation:

The graph helps understand whether financially stable customers are less likely to churn. Customers with lower credit scores may display different banking behavior compared to those with higher creditworthiness.

Members who have medium level credit score are likely to leave bank accounts compared to low and high credit score customers

Tenure-wise Churn Analysis

Churn was analyzed according to the duration of customer relationship with the bank.

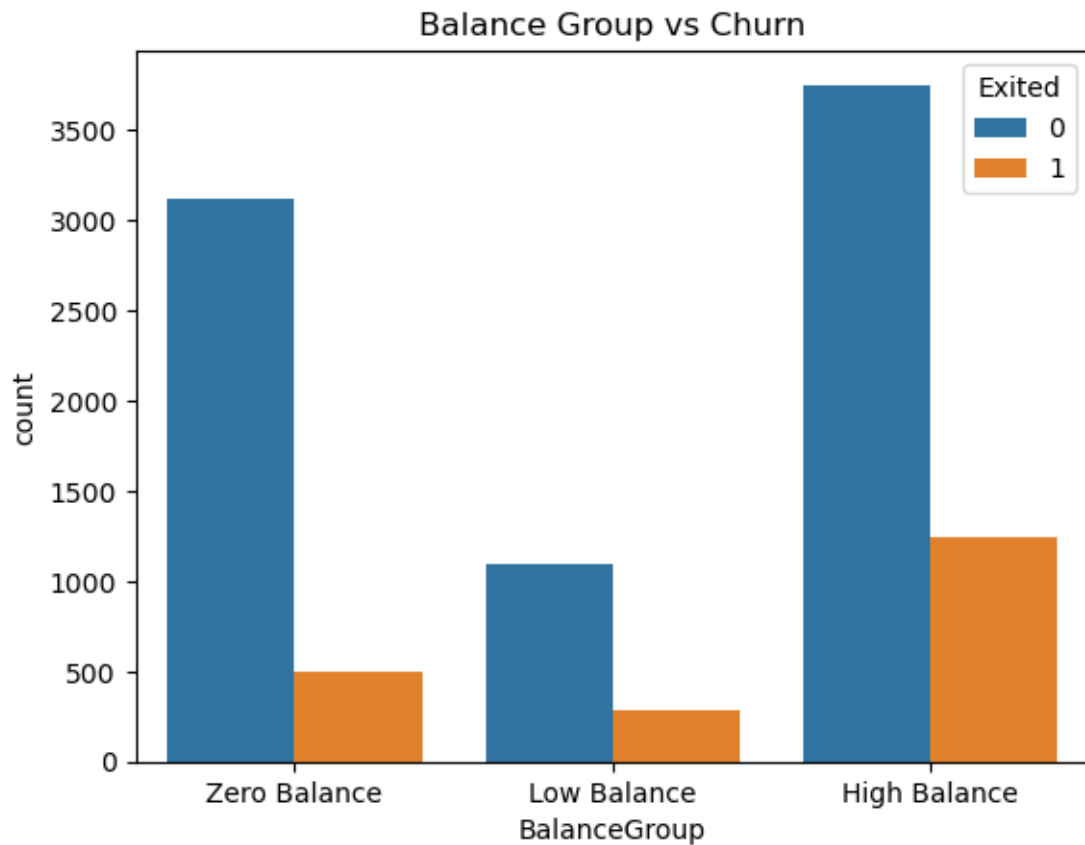


Interpretation:

This analysis helps determine whether newly joined customers are more likely to churn compared to long-term customers. New customers demonstrated relatively higher churn compared to long-term customers.

Balance-wise Churn Analysis

Customers were categorized into balance segments.

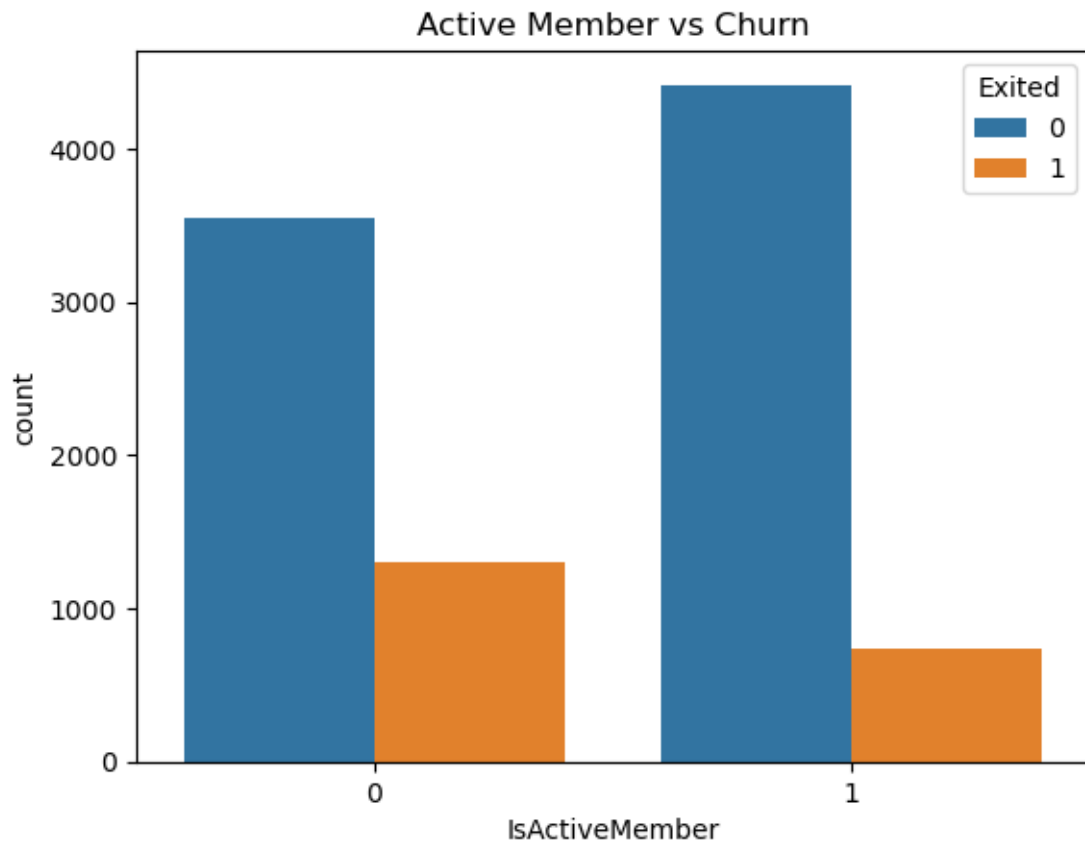


Interpretation:

The analysis identifies whether churn occurs among low-balance customers or financially valuable customers. High-balance customer churn can significantly impact bank revenue.

Active Membership vs Churn

Customer activity level was analyzed to determine its relationship with churn.

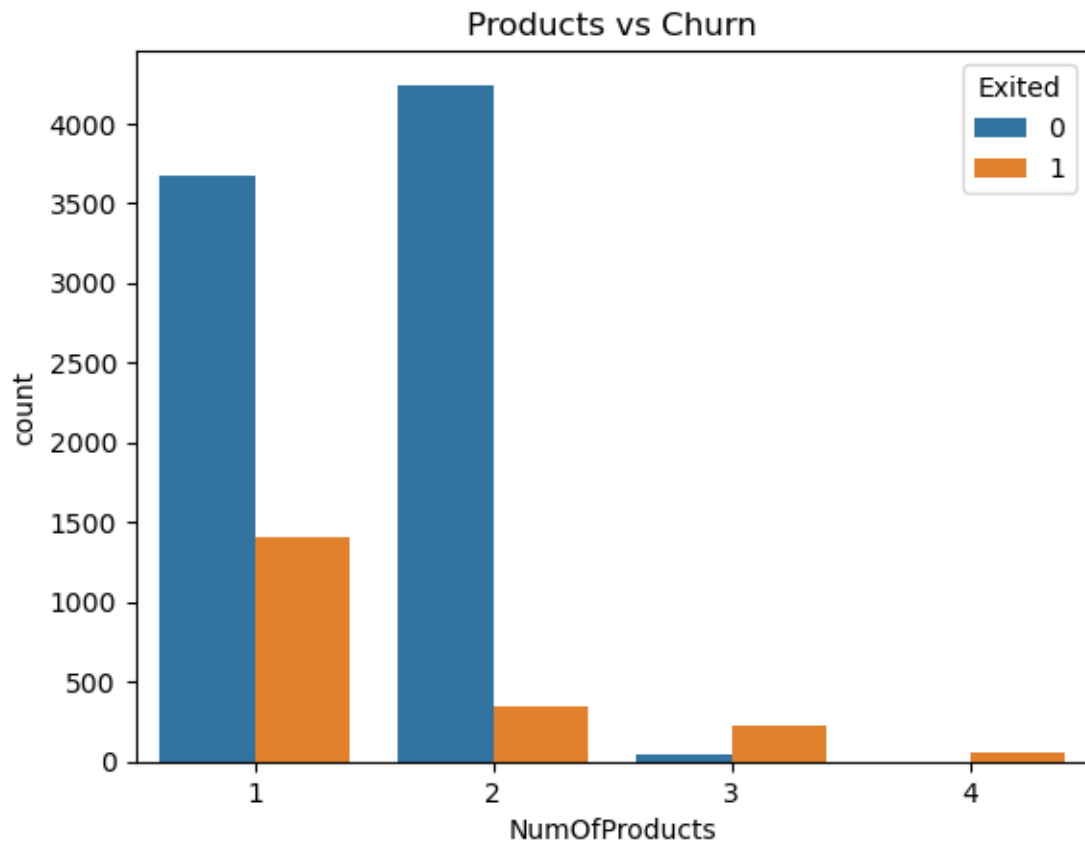


Interpretation:

Inactive customers generally show a higher likelihood of churn. This indicates that customer engagement plays an important role in retention.

Product Usage vs Churn

Churn behavior was analyzed based on the number of products used by customers.



Interpretation:

Customers using fewer banking products may have weaker relationships with the bank, making them more likely to churn.

High-Value Customer Churn Analysis

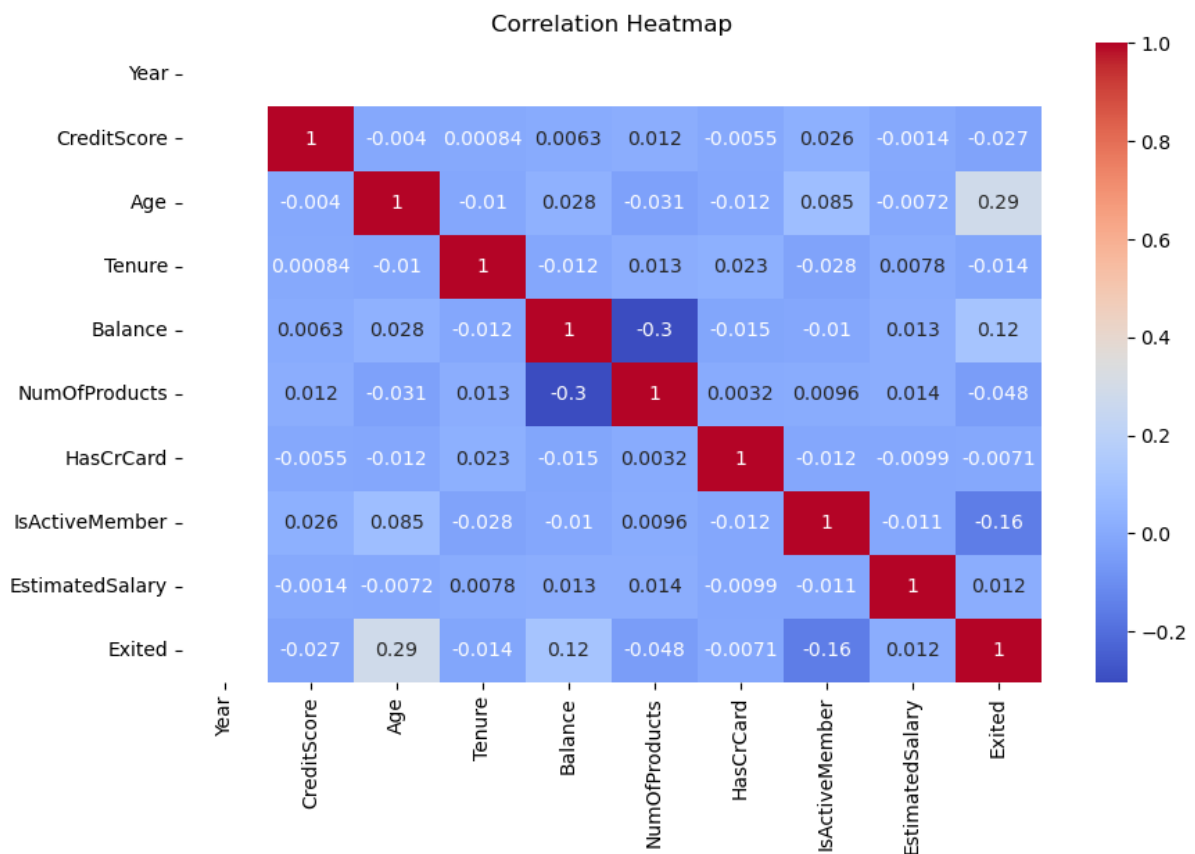
High-value customers were identified using account balance.

Interpretation:

The analysis helps identify whether premium customers are leaving the bank. Churn among high-value customers can create substantial financial risk for the bank.

Correlation Analysis

A correlation heatmap was created to understand relationships between numerical variables and churn.



Interpretation:

The heatmap helps identify variables that have stronger relationships with customer churn. Features such as age, balance, and active membership may show stronger associations with customer exit behavior.

Key Findings from EDA

Based on exploratory data analysis, the following findings were observed:

1. The overall churn rate indicated that a portion of customers exited the bank.
2. Churn differed across countries, with some regions showing higher customer loss.
3. Customer age influenced churn behavior.

Inactive customers showed a higher tendency to churn.

4. Product usage influenced retention, with lower product usage linked to higher churn.
5. Some high-value customers also churned, creating potential revenue risk for banks.
6. Financial and engagement-related variables showed meaningful relationships with churn.

Key Performance Indicators (KPIs)

Key performance indicators can be defined as something which can be counted and compared against some bench mark providing some objective evidence to attain over specific period of time which simply means it's a measure that helps us to know whether we are meeting our goals or not.

In this project Key Performance Indicators (KPIs) are used to measure important customer churn metrics in European banking. These indicators provide a summarized understanding of customer retention, churn behavior, financial profile, and customer engagement levels.

The KPI analysis helps in identifying business risks and understanding the overall customer status of the bank.

KPI calculation process:

Step 1: Dataset Preparation

The cleaned dataset obtained after Exploratory Data Analysis (EDA) was used for KPI calculation.

The dataset included:

- Customer demographics
- Financial information
- Customer engagement variables
- Customer churn indicator (Exited)

The churn variable (Exited) was used as the primary indicator for KPI measurement.

Where:

- Exited = 1 → Customer churned
- Exited = 0 → Customer retained

Step 2: Customer Segmentation for KPI Measurement

Customer segmentation was performed to calculate segment-based churn indicators.

The following segment groups were created:

Segmentation Type Categories

Age Group	Young, Adult, Middle Age, Senior
Credit Score	Low, Medium, High
Tenure	New, Mid-Term, Long-Term
Balance	Zero Balance, Low Balance, High Balance

These segmentations helped identify customer groups with higher churn exposure

KPI Indicators Used

The following KPI indicators were selected for analysis:

KPI Name	Purpose
Overall Churn Rate	Measure total customer exit percentage
Segment Churn Rate	Identify churn among customer segments
High-Value Churn Ratio	Measure churn among premium customers
Geographic Risk Index	Identify regional churn exposure
Engagement Drop Indicator	Compare active and inactive customer churn

Overall Churn Rate

Purpose

To measure the percentage of customers who exited the bank.

Formula:

Overall churn rate= exited customers/total customers *100

Overall Churn Rate: 20.37 %

Interpretation:

This KPI measures total customer attrition within the bank. A higher churn percentage indicates increased customer loss and potential revenue decline.

It shows that 20.37% of customers left the bank.

Segment Churn Rate

Purpose

To identify which customer groups experience higher churn.

Segment churn was calculated for:

- Age Groups
- Credit Score Groups
- Tenure Groups
- Balance Groups

AgeGroup

Young	7.502569
Adult	15.740584
Middle Age	51.123254
Senior	24.784483

Name: Exited, dtype: float64

Credit Score Group

Low	23.639191
Medium	20.285211
High	19.865212

Name: Exited, dtype: float64

Tenure Group

New	20.892626
Mid-Term	19.644645
Long-Term	20.448179

Name: Exited, dtype: float64

Balance Group

High Balance	24.980000
Low Balance	20.824295
Zero Balance	13.823611

Name: Exited, dtype: float64

Interpretation:

Segment churn analysis helps identify vulnerable customer categories. Some customer segments showed higher churn than others, indicating different retention risks.

For age group: from the output given it shows that middle age group people have high churn rate

Credit Score Group Churn Analysis

The credit score churn analysis showed that customers with low credit scores experienced the highest churn rate (23.64%), followed by customers with medium credit scores (20.29%). Customers with high credit scores showed the lowest churn rate (19.87%).

This indicates that customers with lower financial stability were relatively more likely to leave the bank. However, the churn difference across credit score categories was not extremely large, suggesting that credit score alone may not be the strongest factor influencing customer churn.

Tenure Group Churn Analysis

The tenure-based churn analysis showed only small differences in churn across customer relationship periods. New customers had the highest churn rate (20.89%), followed by long-term customers (20.45%), while mid-term customers had the lowest churn rate (19.64%).

This suggests that customer tenure had a relatively moderate impact on churn behavior. Newly joined customers may leave due to dissatisfaction or unmet expectations, while some long-term customers may churn due to changing financial needs or competitive banking alternatives.

Balance Group Churn Analysis – Conclusion

The balance-based churn analysis revealed noticeable differences in customer exit behavior. High-balance customers showed the highest churn rate (24.98%), followed by low-balance customers (20.82%), while zero-balance customers showed the lowest churn rate (13.82%).

This indicates that financially valuable customers were also leaving the bank, which may create significant revenue risk. Since high-balance customers contribute more financial value, retaining these customers should be a priority for the bank.

High-Value Churn Ratio

Purpose

To measure churn among financially valuable customers.

Premium customers were identified using higher account balances.

Formula

High value churn ratio = high value customers churned /total high value customers*100

High Value Churn Ratio: 24.98 %

Interpretation:

This KPI identifies whether premium customers are leaving the bank. High churn among financially valuable customers indicates potential revenue risk.

The analysis revealed that 24.98% of high-value customers churned.

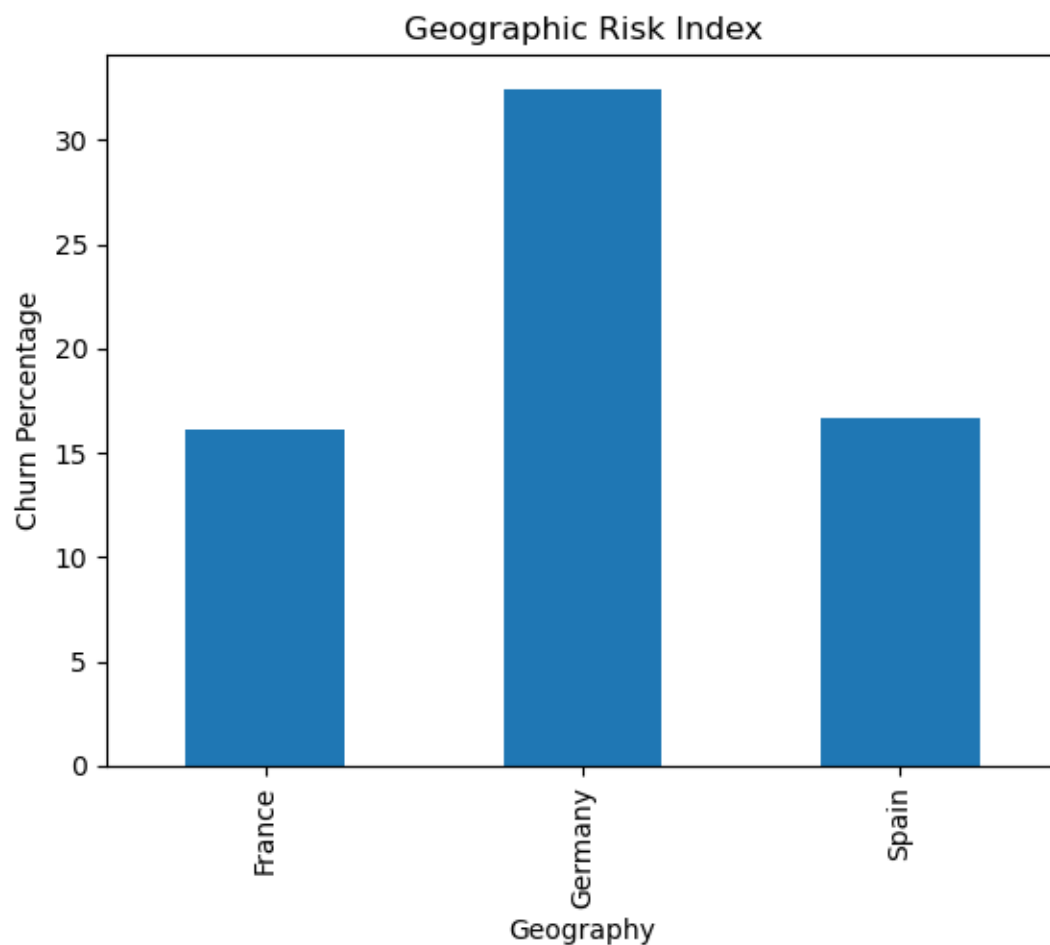
Geographic Risk Index

Purpose

To identify churn exposure across countries.

Country-wise churn percentages were analyzed for:

- France
- Germany
- Spain



Interpretation:

This KPI highlights regional churn risk. Countries with higher churn percentages may require targeted retention strategies. Germany showed relatively higher churn exposure compared to other countries.

Engagement Drop Indicator

Purpose

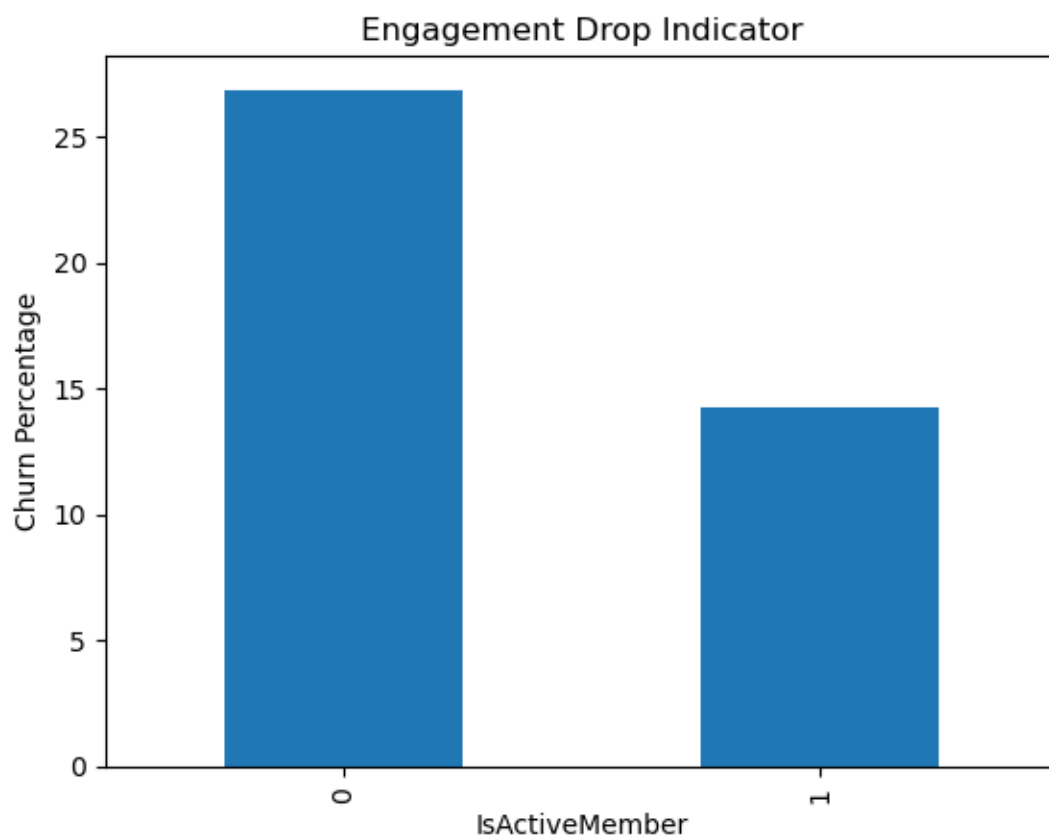
To compare churn between active and inactive customers.

IsActiveMember

0 26.850897

1 14.269074

Name: Exited, dtype: float64



Interpretation:

Inactive customers generally demonstrated higher churn compared to active customers, showing that customer engagement strongly influences retention.

The churn percentage among inactive customers was significantly higher than active customers.

KPI Findings

Based on KPI analysis, the following observations were made:

1. The overall churn rate indicated noticeable customer attrition.
2. Certain customer segments showed higher churn risk.
3. Premium customers also experienced churn.
4. Geographic differences in churn were observed.
5. Customer inactivity showed a strong relationship with churn.

Streamlit Web Application Requirements

Streamlit Dashboard Development

To support interactive customer churn analytics, a Streamlit-based web application dashboard was developed. The dashboard enables users to explore churn patterns across customer segments, demographic groups, and financial profiles in European banking.

The web application was designed to provide dynamic visualizations, KPI tracking, and customer segmentation analysis in an interactive environment.

Core Modules

1. Overall Churn Summary

The Overall Churn Summary module provides a comprehensive overview of customer churn performance across the entire banking dataset.

This module includes:

- Total number of customers
- Number of churned customers
- Number of retained customers
- Overall churn rate percentage

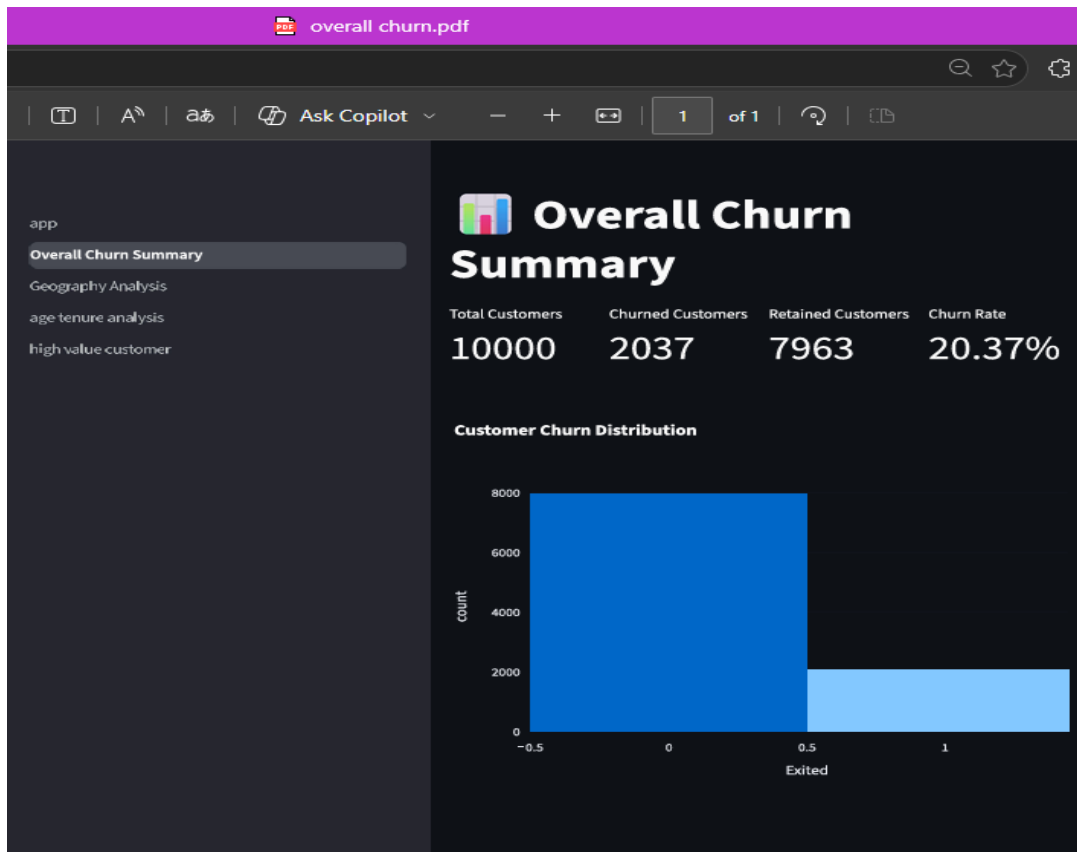
Interactive KPI indicators were developed to summarize customer churn at a glance. Visual charts were included to represent churn distribution between retained and exited customers.

Purpose:

This module helps decision-makers understand the overall level of customer attrition and evaluate the bank's retention performance.

Interpretation:

A higher churn rate indicates increased customer loss and potential business risk. The module helps identify whether customer retention strategies require improvement.



2. Geography-wise Churn Visualization

The Geography-wise Churn Visualization module analyzes churn behavior across different European regions.

The analysis was conducted for:

- France
- Germany
- Spain

This module includes:

- Geography-wise churn comparison
- Regional churn visualization
- Country-level segmentation

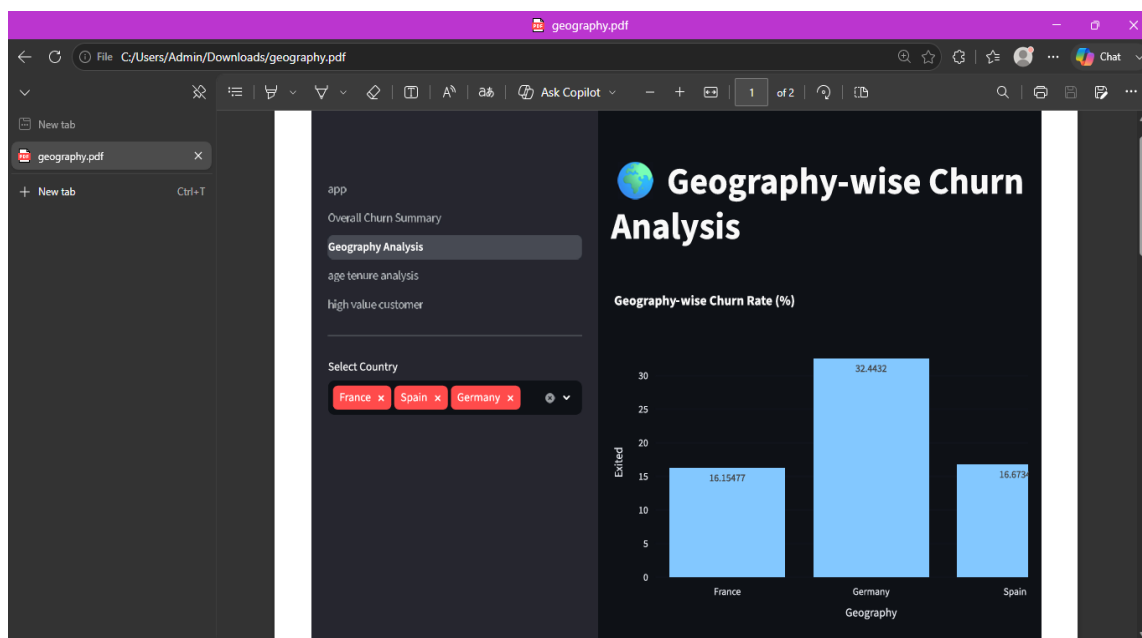
Interactive bar charts were developed to compare churn rates among countries.

Purpose:

This module helps identify geographical regions with higher churn exposure and supports region-specific retention planning.

Interpretation:

Countries with higher churn rates indicate increased customer dissatisfaction or financial instability, requiring focused retention strategies.



3. Age & Tenure Churn Comparison

The Age & Tenure Churn Comparison module evaluates customer churn based on demographic and relationship-duration factors.

The following customer groups were analyzed:

Age Segmentation

- Young Customers (<30 years)
- Adult Customers (30–45 years)

- Middle-Aged Customers (46–60 years)
- Senior Customers (60+ years)

Tenure Segmentation

- New Customers
- Mid-Term Customers
- Long-Term Customers

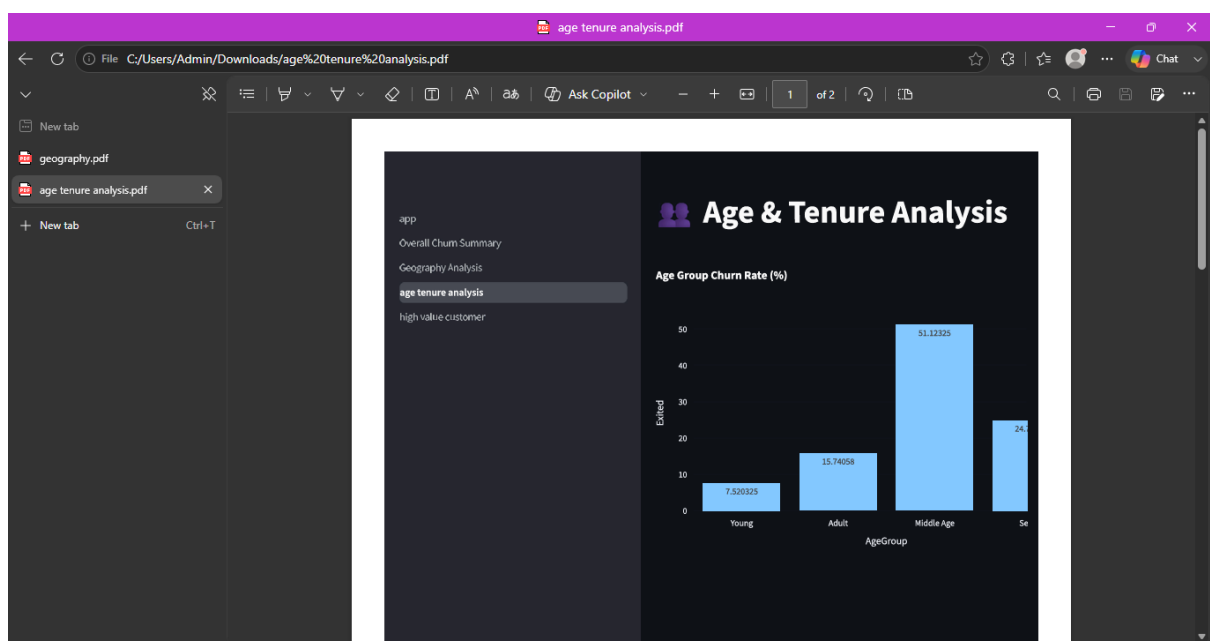
Bar charts were created to compare churn rates across age groups and customer tenure categories.

Purpose:

This module helps identify customer lifecycle stages where churn risk is relatively higher.

Interpretation:

Higher churn in certain age groups or among newer customers suggests the need for better engagement and customer relationship strategies.



4. High-Value Customer Churn Explorer

The High-Value Customer Churn Explorer module focuses on financially valuable customers with higher account balances.

This module includes:

- High-value customer churn ratio
- Balance-based customer segmentation
- Customer engagement comparison
- Inactive vs active customer churn

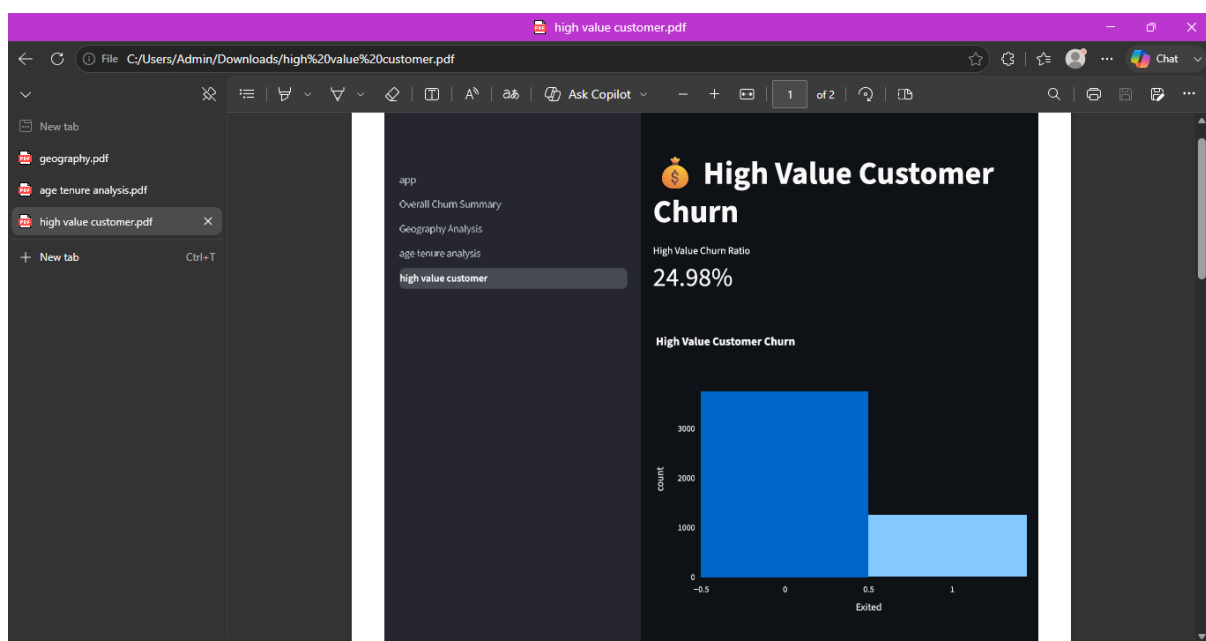
Interactive KPI indicators and charts were implemented to analyze churn among premium customers.

Purpose:

This module helps quantify revenue risk caused by customer attrition among financially significant customer groups.

Interpretation:

Higher churn among premium customers may lead to significant financial losses, making targeted retention strategies essential.



User Capabilities

The Streamlit dashboard provides interactive features that allow users to explore churn patterns effectively.

1. Segment Filters

Users can filter customer data based on selected segments such as geography to analyze churn behavior for specific customer groups.

Purpose:

Segment filtering improves analytical flexibility and enables focused decision-making.

2. Dynamic KPI Updates

The dashboard dynamically updates KPI indicators based on user-selected filters and segmentation conditions.

Examples include:

- Overall churn rate
- High-value customer churn ratio
- Customer retention indicators

Purpose:

Dynamic KPI updates improve real-time analytical understanding and support quick business insights.

3. Drill-Down Views

The dashboard allows users to perform drill-down analysis by moving from overall churn trends to segment-specific customer behavior.

This enables detailed exploration of:

- Geography-wise churn
- Age-group churn
- Tenure-based churn
- High-value customer churn

Purpose:

Drill-down functionality helps decision-makers identify high-risk customer segments and design targeted retention strategies.

Summary

The Streamlit web application successfully transformed customer churn analytics into an interactive and user-friendly dashboard. By integrating KPI indicators, segmentation filters, and visual analytics, the dashboard improved understanding of customer churn patterns across geography, demographics, and financial characteristics in European banking.

User Capabilities

The Streamlit dashboard was developed with interactive user capabilities to improve customer churn analysis and decision-making.

1. Segment Filters

Segment filters were implemented to allow users to analyze customer churn for specific customer groups. The dashboard

enables filtering of customer data based on selected geographical regions.

Users can filter data for:

- France
- Germany
- Spain

This functionality helps users focus on a specific customer segment instead of analyzing the complete dataset.

Purpose:

Segment filters improve analytical flexibility and help decision-makers identify churn patterns within particular customer categories.

Interpretation:

By applying filters, users can compare churn behavior across different customer groups and identify high-risk segments more effectively.

2. Dynamic KPI Updates

Dynamic KPI updates were implemented to automatically update performance indicators based on user-selected filters and customer segments.

The dashboard dynamically updates:

- Overall Churn Rate
- Number of Churned Customers
- Number of Retained Customers

- High-Value Customer Churn Ratio

This ensures that users receive updated analytical insights instantly without manually recalculating values.

Purpose:

Dynamic KPI updates improve real-time monitoring of customer churn and enhance analytical efficiency.

Interpretation:

As users apply filters, KPI values change automatically, helping identify variations in churn performance across different customer segments.

3. Drill-Down Views

Drill-down views were implemented to provide detailed exploration of customer churn behavior.

Users can move from overall churn analysis to more detailed segment-level analysis such as:

- Geography-wise churn patterns
- Age-group churn comparison
- Tenure-based churn analysis
- High-value customer churn exploration

This functionality helps users understand deeper customer churn trends rather than relying only on summary statistics.

Purpose:

Drill-down analysis supports detailed investigation of customer churn patterns and improves strategic decision-making.

Interpretation:

The feature enables identification of high-risk customer categories and supports the development of targeted customer retention strategies.

Overall Conclusion

The project Customer Segmentation & Churn Pattern Analytics in European Banking was carried out to understand customer churn behavior in European banks.

Through customer segmentation, EDA, KPI analysis, and dashboard visualization, the project helped identify how churn differs based on geography, age, tenure, credit score, account balance, and customer activity.

The analysis showed that some customer groups, such as inactive customers, high-balance customers, and customers with lower credit scores, showed relatively higher churn behavior. Geography-wise differences in churn were also observed among different countries.

A Streamlit dashboard was developed to make the analysis more interactive and easy to understand through charts, KPIs, filters, and visualizations.

Overall, this project helped in understanding customer churn patterns and provided useful insights that can support better customer retention strategies in the banking sector.

